

Ratios: Compare, Scale, And Solve

Grades 6-8 visual lesson

Key idea

A ratio compares two quantities in a chosen order. Equivalent ratios keep the same multiplicative relationship.

Today: read ratios, find unit rates, solve proportions, and check answers.

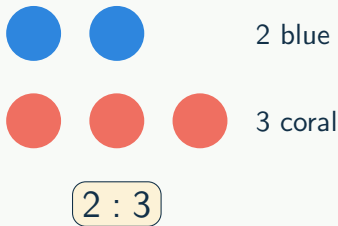


Warm-Up: Read The Comparison

Key idea

The order matters. The ratio of blue counters to coral counters is 2 : 3.

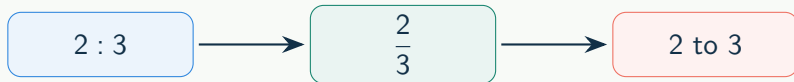
1. Count the blue counters.
2. Count the coral counters.
3. Say the comparison aloud.



One Ratio, Three Ways

Key idea

These forms describe the same ordered comparison.



Always name the quantities: blue to coral, cups to batches, miles to hours.

Equivalent Ratios Keep One Scale Factor

Key idea

Multiply both parts by the same number. Here the scale factor is 2.

$$2 : 3 \xrightarrow{\times 2} 4 : 6$$

Blue	Coral	Ratio
2	3	2 : 3
4	6	4 : 6
6	9	6 : 9

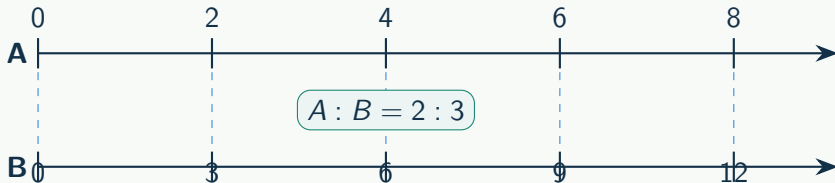
Fast check

Each row simplifies back to 2 : 3.

See The Pattern On A Double Number Line

Key idea

Matching points show the same multiplicative relationship.



Unit Rate Means Per One

Key idea

A car travels 180 miles in 3 hours.
Divide both quantities by 3.

$$\frac{180 \text{ miles}}{3 \text{ hours}} = 60 \text{ miles per hour}$$

180 miles in 3 hours



÷3 on both quantities



60 miles in 1 hour

Set Up A Proportion

Key idea

Keep matching quantities in matching positions.

1. Write the known ratio.
2. Put the unknown in the matching place.
3. Use a scale factor or cross-products.

$$\frac{2 \text{ cups}}{3 \text{ batches}} = \frac{x \text{ cups}}{12 \text{ batches}}$$



$\times 4$



$$x = 2 \times 4 = 8 \text{ cups}$$

Worked Example: Scale A Recipe

Given

2 cups of flour make 3 batches.

Find

How many cups make 12 batches?

Model

$$\frac{2}{3} = \frac{x}{12}$$

Apply

$$3 \times 4 = 12, \text{ so } 2 \times 4 = 8.$$

Result

$x = 8$ cups of flour.

Check

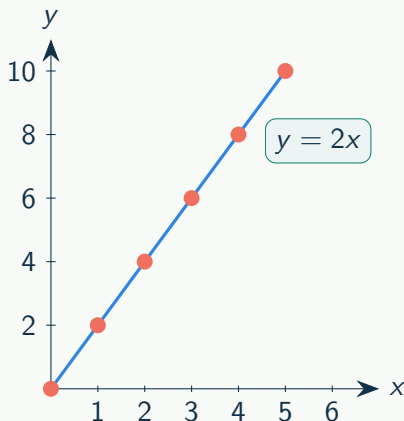
$$\frac{8}{12} = \frac{2}{3} \text{ after dividing by 4.}$$

A Proportional Graph Starts At The Origin

Key idea

For $y = 2x$, every point has the same unit rate, $y/x = 2$.

- ▶ Straight line
- ▶ Passes through $(0,0)$
- ▶ Constant slope



Common Mistake: Changing Only One Part

Incorrect

$2 : 3 \rightarrow 4 : 3$ changes only the first quantity.

$$\frac{2}{3} \neq \frac{4}{3}$$

Accurate

$2 : 3 \rightarrow 4 : 6$ multiplies both quantities by 2.

$$\frac{2}{3} = \frac{4}{6}$$

One scale factor must act on both quantities.

Your Turn

1. Unit rate

A cyclist travels 42 miles in 3 hours. Find miles per hour.

2. Missing value

$$\frac{4}{7} = \frac{x}{21}$$

3. Proportional or not?

Does $y = 3x + 1$ describe a proportional relationship? Explain.

Hints

Divide to get per one. Look for a scale factor. Check whether the graph passes through the origin.

Check Your Reasoning

1. Unit rate

$42 \div 3 = 14$ miles per hour.

2. Missing value

$7 \times 3 = 21$, so $4 \times 3 = 12$ and $x = 12$.

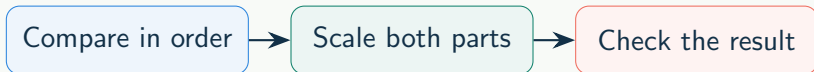
3. Proportional or not?

No. The $+1$ means the graph does not pass through $(0, 0)$.

Key idea

A strong answer names the relationship and checks it using numbers, a table, an equation, or a graph.

Quick Review



Remember

Equivalent ratios have one constant multiplier. A unit rate compares to one. A proportional graph is a straight line through the origin.

Ready for the scored quiz?